University of Illinois at Chicago
Department of Chemical Engineering

SAFETY QUIZ: CHE382-Chemical Engineering Laboratory II

Please answer the following questions using the information from the 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases given in Safety Assignment I. Note: Answers from another source may not be considered correct. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the definition of a toxic chemical?

A toxic chemical is any substance which when inhaled, ingested, injected or absorbed through the skin can cause damage to the biological structures or interfering with normal bodily functions.

2. Name the 4 types of toxicity damage from a toxic chemical:

   Acute toxicity
   Local toxicity
   Chronic toxicity
   Systemic toxicity

3. Name two parameters that can affect toxicity:

   Mode of Entry
   Duration of exposure

4. What is the PEL (Permissible Exposure Limit)?

PEL is the maximum concentration of a chemical that workers may be exposed to during normal working conditions so no harmful effects could occur to them.

5. What is the DOT (Department of Transportation) definition of compressed gas?

A compressed gas is a gas stored under pressure in a container including liquefied gases and is considered a hazard because of the stored pressure which has the potential to be a hazard if it gets released quickly

6. a) How does one determine how much material remains in a non-liquified gas cylinder?

To determine how much material remains in a non-liquified gas cylinder we can read the cylinder pressure gauge and relate the pressure to the known cylinder volume

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The high-pressure gauge on the two-stage regulator

7. How does one determine how much material remains in a liquified gas cylinder?

To determine the amount of material we can weigh the cylinder and subtract it to the empty weight, since the pressure is constant until the liquid is gone

8. Why do insidious hazards represent a problem to workers?

Because it takes time to see symptoms and can cause long term or delayed heath effects

9. What are the four general categories of insidious hazards?

Carcinogens
Teratogens
Reproductive toxins
Mutagens

10. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

Using Proper ventilation
Using personal protective equipment which is also known as PPE

11. What is the definition of a reactive chemical?

A reactive chemical is a substance that can change chemically due to heat, pressure, or contact with other materials that might cause a reaction.

12. Name the two groups and their sub-headings used to categorize reactive chemicals:

Water reactive chemicals
Air reactive chemicals

13. List two precautionary measures for handling and/or storing reactive chemicals.

Control exposure to heat, ignition sources, and moisture
Store away from materials that could potentially cause a reaction

14. How are corrosive chemicals generally defined?

Corrosive chemicals are substances that cause irreversible damage to living tissue or material via chemical means

15. How do corrosive chemicals act on body tissues?

    By damaging tissues either through a chemical reaction, burn or permanent damage

16. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

    True

17. Name the six categories of common corrosive liquids.

    Strong acids
    Weak acids
    Strong bases
    Weak bases
    Oxidizers
    Dehydrating agents

18. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

    Strong bases because of the high amount of use in the industry

19. Name the three components that make up the FIRE TRIANGLE:

    Fuel
    Oxidizer
    Ignition source

20. What category of flammables is the most common source of fires in industry?

    Flammable solvents